

# Insulin Dose Calculator — Parameter Grid (Ranges)

## ORDER ENTRY — HARD VALIDATION (ALLOWED RANGES)

| Parameter               | U-100 Standard (Units)         | U-10 Dilute (Syringe Units) |
|-------------------------|--------------------------------|-----------------------------|
| Insulin : carb ratio    | 1–100 g/Unit                   | 1.5–40 g/Syr Unit           |
| Correction factor (ISF) | 4–300 mg/dL/Unit               | 10–200 mg/dL/Syr Unit       |
| Target blood glucose    | 120–300 mg/dL                  | 120–300 mg/dL               |
| Rounding increment      | 0.5 or 1 Unit (prescriber-set) | 1 Syr Unit (fixed)          |

## ORDER SIGN — DISCERN RULE ALERTS (SOFT LIMITS, BY AGE)

| Age band       | U-100 Standard only |                           | Both U-100 & U-10   |                   |
|----------------|---------------------|---------------------------|---------------------|-------------------|
|                | I:C low (g/Unit)    | I:C high (g/Unit)         | CF (mg/dL/Unit)     | Target BG (mg/dL) |
| 0–5 y          | below 10            | 100 or more<br>(all ages) | below 50 · over 200 | under 130         |
| 5–11 y         | below 4             |                           | below 15 · over 200 | over 250          |
| 12 y and older | below 2             |                           | over 200            | over 250          |

## CALCULATOR — GUARDS & REINFORCEMENT

| Parameter / input          | U-100                                                                                                    | U-10                             |
|----------------------------|----------------------------------------------------------------------------------------------------------|----------------------------------|
| Target BG · < 5 y          | 130–300 mg/dL                                                                                            | 130–300 mg/dL                    |
| Target BG · 5–11 y         | 120–250 mg/dL                                                                                            | —                                |
| Target BG · 12 y and older | 120–250 mg/dL                                                                                            | —                                |
| Carbohydrates eaten        | 0–300 g; Alerted Calculation cap of 250 g                                                                | 0–99 g                           |
| Current blood glucose      | 1–2000 mg/dL; alerted cap at 600                                                                         | 1–2000 mg/dL; alerted cap at 600 |
| Correction factor · ≥ 12 y | verify > 150 mg/dL/Unit — reinforced at calculator                                                       | —                                |
| Insulin on board (IOB)     | Optional checkbox; subtracts still-active insulin from prior doses within a 2 h lookback (anti-stacking) | same                             |

## BLOOD-GLUCOSE DECISION THRESHOLDS (both concentrations)

| Condition             | Behavior                                                                  |
|-----------------------|---------------------------------------------------------------------------|
| <b>BG = 0</b>         | Critical hypoglycemia — functionally never triggered (entry minimum is 1) |
| <b>BG &lt; 70</b>     | Hypoglycemia — all dosing suppressed                                      |
| <b>BG &gt; target</b> | Correction applies (otherwise prandial only)                              |
| <b>BG &gt; 600</b>    | Correction calculated at 600 cap; prandial unaffected                     |

- Order details are hard limited at “add to scratch pad” and further refined via soft advisory warnings at “sign order” when detail values for ICR, CF, or target BG are allowed but outside common values for age groups.
- Target BG ranges are age tiered and concentration-independent: the same soft order-warn (on sign) and calculator limits apply to U-100 and U-10 at each tier.
- The I:C high advisory fires at 100 (the maximum), flagging attempts to use the calculator for sliding-scale-style orders where a very high ratio makes prandial coverage negligible.
- CF high warn (over 200, all ages) fires as an order-sign Discern rule. The CF verify cue (over 150, ≥ 12 y) is not an order-sign rule — it is reinforced in the calculator only.
- Ordered values that are outside soft warning limits are re-checked at the calculator.
- BG entry is 1–2000 mg/dL; correction uses up to 600 (alerted). BG = 0 cannot be entered, so critical hypoglycemia is functionally never triggered at the calculator.
- Out-of-range parameters are corrected up or down to the nearest violated boundary.
- Reference only — not for clinical use without pharmacy & informatics committee review.